krishnasree08@krishnasree08

```
 GNU nano 7.2
#ifndef OPERATIONS_H
#define OPERATIONS_H

int add(int a, int b);
int subtract(int a, int b);

#endif
```

krishnasree08@krishnasree08

```
 GNU nano 7.2                                o
#include "operations.h"

int add(int a, int b) {
    return a + b;
}

int subtract(int a, int b) {
    return a - b;
}
```

krishnasree08@krishnasree08

```
 GNU nano 7.2                                main.c *
#include <stdio.h>
#include "operations.h"

int main(void) {
    printf("Addition: %d\n", add(10, 5));
    printf("Subtraction: %d\n", subtract(10, 5));
    return 0;
}
```

```
krishnasree08@krishnasree08:~$ mkdir -p ~/ossp_session21_22
cd ~/ossp_session21_22
krishnasree08@krishnasree08:~/ossp_session21_22$ nano operations.h
krishnasree08@krishnasree08:~/ossp_session21_22$ nano operations.c
krishnasree08@krishnasree08:~/ossp_session21_22$ nano main.c
krishnasree08@krishnasree08:~/ossp_session21_22$ gcc -Wall -Wextra main.c op
erations.c -o modules
./modules
Addition: 15
Subtraction: 5
krishnasree08@krishnasree08:~/ossp_session21_22$ nano error_logging.c
krishnasree08@krishnasree08:~/ossp_session21_22$ gcc -Wall -Wextra error_log
ging.c -o error_logging
./error_logging
File error: No such file or directory
Error code: 2
krishnasree08@krishnasree08:~/ossp_session21_22$ gcc -Wall -Wextra error_log
ging.c -o error_logging
./error_logging
File error: No such file or directory
Error code: 2
```

Overall Result – Sessions 21 & 22

Thus, the programs were implemented to demonstrate modular programming, runtime-error detection and logging, dynamic memory allocation and deallocation, memory-leak detection using Valgrind, and automated testing using shell scripts. The experiments helped improve code reliability, error handling, memory management, and software testing.